

RGPV | B.Tech First Year

ENGINEERING PHYSICS

(BT-201)

UNIT 3 — Handwritten Notes

Semiconductor Physics & Band Theory

☐ Topics Covered in this Unit

1. *Kronig-Penney Model*
2. *PN Diode*
3. *Avalanche vs Zener Breakdown*
4. *Intrinsic vs Extrinsic Semiconductor*
5. *Fermi Level & Bloch's Theorem*
6. *Solar Cell*
7. *Zener Diode*
8. *Hall Effect*

📖 HOW TO USE THESE NOTES

Blue ink = explanation | Black = headings | Yellow box = exam-important | Green dotted box = trick/mnemonic

UNIT 3

1. Kronig–Penney Model

Why do we need this model?

In a real crystal, an electron does not move freely. It moves inside a **periodic potential** created by the regularly arranged positive ions of the lattice. The Kronig–Penney (K–P) model is a simple, solvable model that shows **why electron energies in a solid split into allowed "bands" and forbidden "gaps."** This is the starting point of the whole Band Theory of Solids.

DEFINITION

The Kronig–Penney model treats a crystal as a one-dimensional array of finite rectangular potential wells (atoms) separated by potential barriers (space between atoms), repeating periodically with period $(a+b)$.

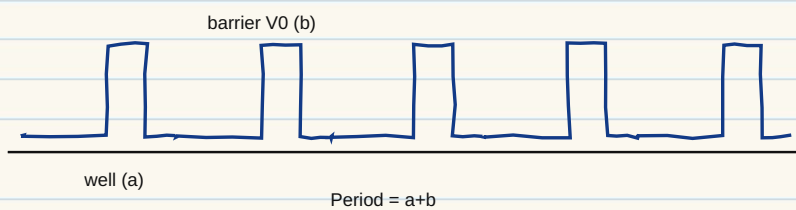


Fig 1.1 — Periodic square-well potential used in K-P model

Basic idea (in simple words)

- Electron is free inside the well (low potential, width " a ") → behaves like a free particle.
- Between wells there is a barrier of height V_0 and width " b " → electron can still "tunnel" through it (quantum mechanics).
- We solve Schrodinger's equation separately in the well and the barrier, then match the wave function and its slope at each boundary (using Bloch's theorem, see Topic 5).

The famous K-P result:

$$P (\sin \alpha a / \alpha a) + \cos \alpha a = \cos(\kappa a)$$

where $\alpha^2 = 2mE/\hbar^2$, $P = (mV_0ba)/\hbar^2$ is called the 'scattering power' of the barrier , κ = electron wave vector

What this equation tells us

- Left-hand side can take **any value** as α changes, but the right side, $\cos(\kappa a)$, can only lie between **-1 and +1**.
- So the equation has a solution (i.e. an allowed electron state) **only for certain ranges of energy E**.
- These allowed ranges are called **ENERGY BANDS**; the forbidden ranges in between are **FORBIDDEN GAPS** (Band Gaps).

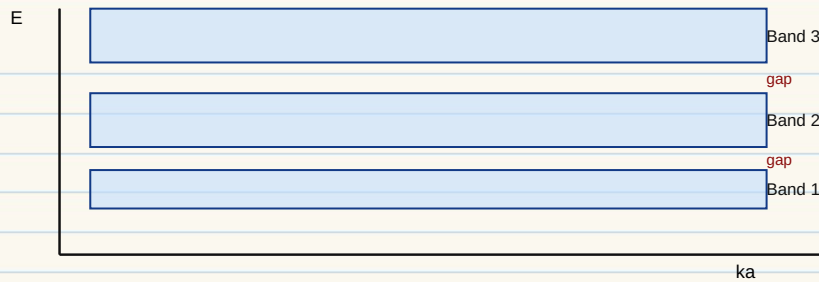


Fig 1.2 — Allowed energy bands separated by forbidden gaps (E vs ka)

Special / limiting cases (often asked)

Case	Condition	Result
$P \rightarrow \infty$	Barrier very strong	Bands become infinitely narrow $\rightarrow$ discrete levels (isolated atom)
$P \rightarrow 0$	Barrier vanishes	Electron is completely free $\rightarrow$ continuous energy (no gaps)

★ EXAM IMPORTANT

1) Derivation/statement of the K-P equation. 2) Significance of "P". 3) Explain origin of energy bands and how this explains conductors, insulators & semiconductors. (Very frequently asked, 7 marks)

🔑 TRICK TO REMEMBER

"P Sin Cos Cos" $\rightarrow P(\sin \alpha a / \alpha a) + \cos \alpha a = \cos(\kappa a)$. Remember: **bigger P = more tightly bound electron = narrower bands**.

✓ QUICK REVISION

- K-P model = periodic square wells + barriers (1-D crystal model)
- Gives relation: $P(\sin \alpha a / \alpha a) + \cos \alpha a = \cos \kappa a$
- Because RHS is limited to $[-1, +1]$, energy splits into **allowed bands** & **forbidden gaps**
- $P \rightarrow \infty$ gives isolated atom levels; $P \rightarrow 0$ gives free electron (no gaps)

2. PN Diode

DEFINITION

A PN diode is a two-terminal semiconductor device formed by joining a P-type and an N-type semiconductor. It allows current to flow easily in one direction only.

Formation of the junction

- Near the junction, free electrons from the N-side diffuse into the P-side and holes from P-side diffuse into N-side.
- This leaves behind fixed (immobile) ions: **negative ions on P-side, positive ions on N-side** near the junction.
- This region, empty of free charge carriers, is called the **Depletion Region**.
- The exposed ions create an internal electric field / potential barrier called **Barrier Potential (V_0)** $\approx 0.7\text{ V}$ for Silicon, $\approx 0.3\text{ V}$ for Germanium, which opposes further diffusion.

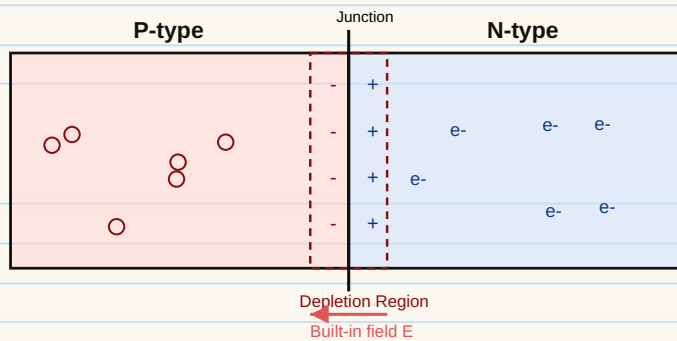


Fig 2.1 — PN junction showing depletion region and built-in field

Biasing

Forward Bias	Reverse Bias
+ve terminal $\rightarrow$ P side, -ve $\rightarrow$ N side	+ve terminal $\rightarrow$ N side, -ve $\rightarrow$ P side
Depletion region becomes <i>thin</i>	Depletion region becomes <i>wide</i>
Barrier potential decreases	Barrier potential increases
Large current flows (mA)	Only tiny leakage current (μA) flows

Diode current equation

$$I = I_0 (e^{qV/\eta kT} - 1)$$

I_0 = reverse saturation current, q = electron charge, k = Boltzmann constant, T = temperature (K), η = ideality factor (1 for Ge, 2 for Si)

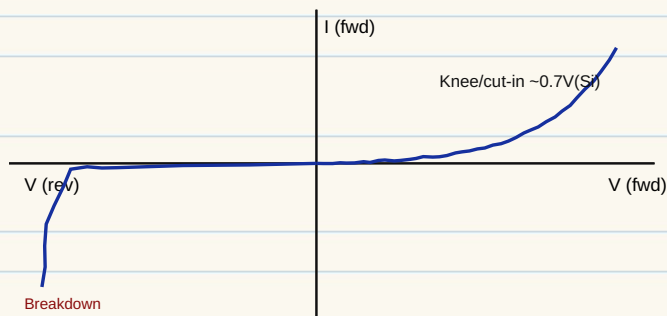
V-I Characteristics

Fig 2.2 — Typical diode V-I characteristic curve

- **Cut-in / knee voltage:** voltage below which forward current is negligible (0.7V Si, 0.3V Ge)
- After the knee, current rises **exponentially** with small increase in voltage.
- In reverse bias, current stays almost constant (I_0) until **breakdown voltage** is reached (see Topic 3).

★ EXAM IMPORTANT

Diode equation derivation, V-I characteristics diagram with labelled knee voltage, and difference between ideal & practical diode are frequently asked (previous year favourite topic).

🧠 TRICK TO REMEMBER

'FIN' — Forward = thin depletion region (low resistance); **'RIW'** — Reverse = wide depletion region (high resistance).

✓ QUICK REVISION

- PN junction = P + N semiconductor joined together; creates depletion region + barrier potential
- Forward bias → thin depletion, large current; Reverse bias → wide depletion, tiny current
- $I = I_0(e^{qV/\eta kT} - 1)$
- Cut-in voltage: 0.7V (Si), 0.3V (Ge)

3. Avalanche vs Zener Breakdown

When reverse voltage on a diode is increased beyond a limit, the diode suddenly starts conducting heavily. This is called **Reverse Breakdown**. There are two mechanisms:

(a) Zener Breakdown

- Occurs in **heavily doped** junctions → depletion region is very **thin**.
- A thin region + reverse voltage gives a very strong electric field ($\sim 10^7$ V/m) even at low voltage.
- This strong field directly **breaks (ruptures) covalent bonds** and pulls valence electrons free — this is called **internal field emission / quantum tunnelling**.
- Occurs at **low reverse voltages, typically below 6 V**.

(b) Avalanche Breakdown

- Occurs in **lightly doped** junctions → depletion region is **wide**.
- Minority carriers crossing the wide depletion region are accelerated by the reverse field and gain high kinetic energy.
- These fast carriers collide with atoms and **knock out** more electron-hole pairs — each of those again collide & multiply → a **chain reaction ("avalanche") of carrier multiplication**.
- Occurs at **higher reverse voltages, typically above 6 V**.

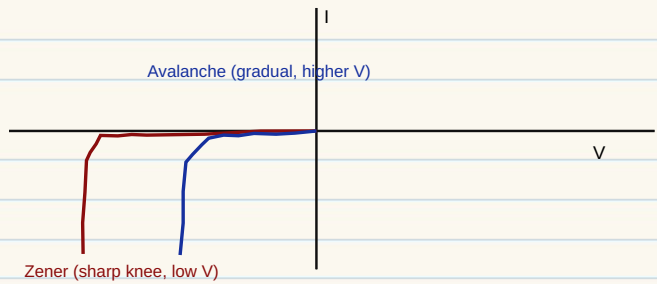


Fig 3.1 — Comparison of Zener (sharp) and Avalanche (gradual) reverse breakdown

Property	Zener Breakdown	Avalanche Breakdown
Doping level	Heavy	Light
Depletion width	Thin	Wide
Mechanism	Direct rupture of bonds by strong field	Carrier multiplication by collision
Breakdown voltage	Low (< 6 V)	High (> 6 V)
Temp. coefficient	Negative	Positive

★ EXAM IMPORTANT

"Differentiate Zener and Avalanche breakdown" is a classic 5-mark question — write the table above directly. Also remember temperature coefficient sign (Zener: negative, Avalanche: positive).

🔑 TRICK TO REMEMBER

"**Z** for **Z**ero-effort" — Zener breaks bonds directly (no collision needed), happens at **LOW** voltage. "**A** for **A**ccident **chain**" — Avalanche = chain of collisions, needs **HIGH** voltage & wide path to build up speed.

✓ QUICK REVISION

- Both are reverse breakdown mechanisms in a diode
- Zener: heavy doping, thin depletion, direct field breaking of bonds, <6V, negative temp coeff.
- Avalanche: light doping, wide depletion, carrier multiplication by collision, >6V, positive temp coeff.

4. Intrinsic vs Extrinsic Semiconductor

Intrinsic Semiconductor

DEFINITION

A pure semiconductor (like pure Silicon or Germanium) with no impurity added. Number of free electrons = Number of holes.

- At 0 K it behaves like a perfect insulator (valence band full, conduction band empty).
- At room temperature, thermal energy breaks a few covalent bonds → equal electron-hole pairs (EHP) are generated.
- Electron concentration n = Hole concentration $p = n_i$ (intrinsic concentration)

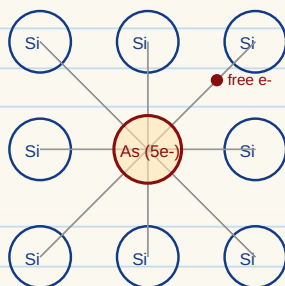
Extrinsic Semiconductor

DEFINITION

A semiconductor with a small controlled amount of impurity (doping) added to increase its conductivity.

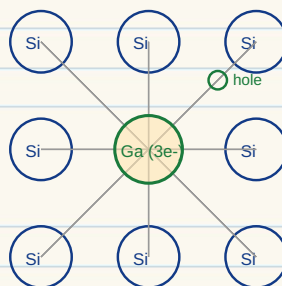
N-type

- Doped with **Pentavalent** impurity (P, As, Sb) — 5 valence electrons.
- 4 electrons form bonds, 5th electron is **free** → donor atom.
- Electrons = Majority carriers, Holes = Minority carriers.



P-type

- Doped with **Trivalent** impurity (B, Al, Ga, In) — 3 valence electrons.
- Only 3 bonds complete, 1 bond incomplete → creates a **hole** → acceptor atom.
- Holes = Majority carriers, Electrons = Minority carriers.



Mass Action Law (very important formula)

$n \times p = n_i^2$

This holds true for both intrinsic and extrinsic semiconductors at thermal equilibrium (product of electron and hole concentration is constant at a given temperature).

Property	Intrinsic	Extrinsic
Purity	Pure, no impurity	Doped with impurity
Carrier conc.	$n = p = n_i$	$n \neq p$
Conductivity	Low	High (controllable)
Example	Pure Si, Ge	Si/Ge + Dopant (P, B etc.)

★ EXAM IMPORTANT

Difference table (above), mass-action law $n \cdot p = n_i^2$, and diagram of doped lattice with donor/acceptor levels are all frequently asked short/long questions.

🔑 TRICK TO REMEMBER

“PEN”TA gives N, “TRI” gives P — PEN(TA)valent (5) → N-type (N)egative/extra electron); TRIvalent (3) → P-type (P)ositive/hole).

✔ QUICK REVISION

- Intrinsic = pure, $n = p = n_i$
- Extrinsic = doped; N-type from pentavalent (donor), P-type from trivalent (acceptor)
- Mass action law: $n \times p = n_i^2$ (always true at equilibrium)

5. Fermi Level & Bloch's Theorem

Fermi Level (E_F)

DEFINITION

The Fermi level is the energy level at which the probability of finding an electron is exactly 50% at a given temperature. It is a reference energy level, not necessarily a real, occupied state.

Fermi-Dirac probability function

$$f(E) = 1 / (1 + e^{(E-E_F)/kT})$$

Position of Fermi level

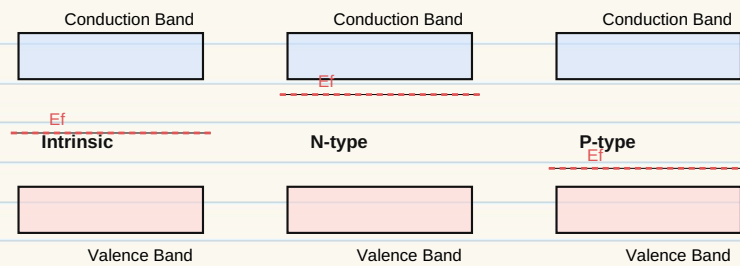


Fig 5.1 — Fermi level position: intrinsic (middle), N-type (near CB), P-type (near VB)

- **Intrinsic semiconductor:** E_F lies almost exactly in the *middle* of the forbidden gap.
- **N-type:** E_F shifts *close to the Conduction Band* (more electrons available).
- **P-type:** E_F shifts *close to the Valence Band* (more holes available).
- As doping increases, E_F moves further towards that band.

Bloch's Theorem

DEFINITION

Bloch's theorem states that the wave function of an electron moving in a periodic potential (like inside a crystal lattice) can be written as a plane wave modulated by a function that has the same periodicity as the lattice.

$$\Psi(x) = u_K(x) \cdot e^{ikx}$$

where $u_K(x + a) = u_K(x)$, i.e. $u_K(x)$ repeats with the same period 'a' as the lattice. e^{ikx} is a plane travelling wave part.

Why it matters

- It proves that an electron in a perfectly periodic crystal can move through it like a free particle (with a modified/"effective" mass), instead of being scattered by every atom.
- It is the mathematical foundation used to derive the Kronig-Penney result (Topic 1) and the whole band theory of solids.
- Functions of this type are called **Bloch functions** and the electrons obeying them are called **Bloch electrons**.

★ EXAM IMPORTANT

Statement + equation of Bloch's theorem, and 'position of Fermi level in intrinsic/N/P type semiconductor with diagram' — both are repeatedly asked (2-mark and 7-mark both types).

🔑 TRICK TO REMEMBER

"N is Near top, P is near Pit(bottom)" — N-type Fermi level near top (Conduction Band), P-type Fermi level near bottom (Valence Band).

✓ QUICK REVISION

- Fermi level = energy with 50% occupation probability
- $f(E) = 1/(1 + e^{(E - E_F)/kT})$
- E_F : middle (intrinsic), near CB (N-type), near VB (P-type)
- Bloch's theorem: $\Psi(x) = u(x) e^{ikx}$, $u(x)$ has lattice periodicity — basis of band theory

6. Solar Cell

DEFINITION

A solar cell is a PN junction device that converts light energy directly into electrical energy using the *Photovoltaic Effect*.

Working principle

- Sunlight (photons) falls on the thin N-side and penetrates to the junction.
- If photon energy $\geq$ band gap energy (E_g), it generates an **electron-hole pair (EHP)** near the junction.
- The built-in electric field of the depletion region **separates** these carriers — electrons pushed to N-side, holes pushed to P-side.
- This separation of charge creates a potential difference → if an external circuit is connected, current flows — this is the **Photovoltaic (PV) effect**.

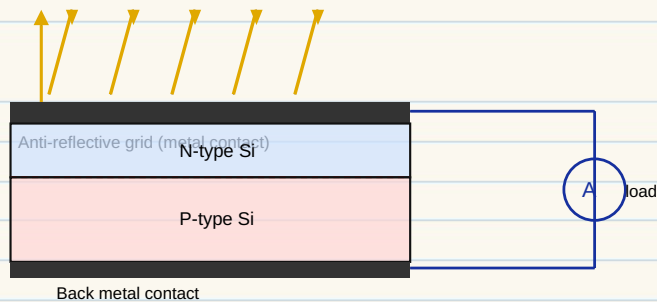


Fig 6.1 — Structure of a silicon solar cell

Important parameters

Term	Meaning
I_{SC}	Short-circuit current (max current, at $V=0$)
V_{OC}	Open-circuit voltage (max voltage, at $I=0$)
Fill Factor (FF)	$FF = P_{max} / (V_{OC} \times I_{SC})$
Efficiency (η)	$\eta = (P_{max} / P_{in}) \times 100\%$

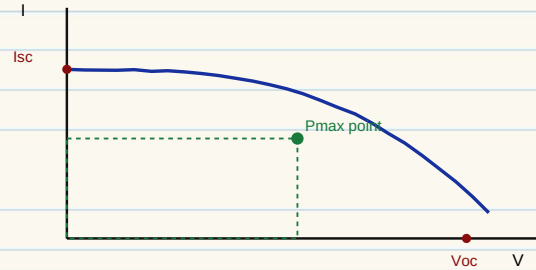


Fig 6.2 — I-V characteristic curve of a solar cell showing max power point

$$\eta = (V_{OC} \times I_{SC} \times FF) / P_{in} \times 100\%$$

★ EXAM IMPORTANT

Photovoltaic effect explanation, labelled solar cell diagram, and formulas of Fill Factor & Efficiency are exam favourites (numerical problems also asked using these formulas).

🔑 TRICK TO REMEMBER

'Light In, Current Out' — remember solar cell = PN junction working in reverse of an LED (LED: electricity → light; Solar cell: light → electricity).

✓ QUICK REVISION

- Solar cell = PN junction using Photovoltaic Effect (light → electricity)
- Photon energy $\geq E_g$ needed to create EHP; junction field separates carriers
- Key terms: I_{sc} , V_{oc} , Fill Factor, Efficiency $\eta = P_{max}/P_{in} \times 100\%$

7. Zener Diode

DEFINITION

A Zener diode is a heavily-doped PN junction diode specially designed to operate continuously in the **reverse breakdown region** without getting damaged, and is mainly used as a **voltage regulator**.

Key features

- Symbol has a bent line (Z-shaped) at the cathode, unlike a normal diode.
- Normally operated in **reverse bias**.
- Once breakdown (Zener voltage V_Z) is reached, voltage across it stays **almost constant** even if current through it changes a lot — this property is used for voltage regulation.
- A series resistor R_S is always used to limit current and protect the diode.

Zener as Voltage Regulator

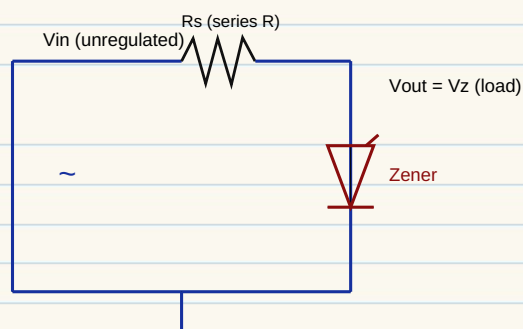


Fig 7.1 — Simple Zener voltage regulator circuit

- Unregulated input V_{in} is applied through series resistor R_S .
- Zener diode is connected in reverse, parallel with the load.
- As long as V_{in} stays above V_Z , output across load stays fixed at V_Z , even if V_{in} or load current varies.

Normal (Rectifier) Diode	Zener Diode
<i>Operated in forward bias</i>	<i>Operated in reverse breakdown region</i>
<i>Lightly doped</i>	<i>Heavily doped</i>
<i>Breakdown is destructive</i>	<i>Breakdown is normal working condition</i>
<i>Used for rectification</i>	<i>Used for voltage regulation, reference voltage</i>

Applications

- *Voltage regulator circuits*
- *Reference voltage source*
- *Over-voltage / surge protection ("clipper" circuits)*
- *Meter protection circuits*

★ EXAM IMPORTANT

Zener as voltage regulator circuit diagram + working explanation is a guaranteed question. Also remember: Zener diode symbol and V - I characteristic (link with Topic 3).

💡 TRICK TO REMEMBER

"Zener = Zaroori Voltage constant" (Hindi trick) — Zener diode's whole job is to keep output voltage constant/fixed, that's why it's used as a regulator.

✔ QUICK REVISION

- *Zener diode = heavily doped diode meant to work in reverse breakdown*
- *Keeps output voltage constant (V_Z) → used as voltage regulator*
- *Needs series resistor R_S to limit current*

8. Hall Effect

DEFINITION

When a current-carrying conductor/semiconductor is placed in a magnetic field perpendicular to the direction of current, a voltage develops across the material in a direction perpendicular to both the current and the field. This is called the **Hall Effect**, and the voltage produced is the **Hall Voltage (V_H)**.

Why it happens (simple explanation)

- Current I flows through the slab; magnetic field B is applied perpendicular to it (into/out of the page).
- Moving charge carriers experience a sideways force — the **Lorentz force**, $F = qv \times B$.
- This pushes carriers to one side of the slab, creating a charge imbalance (one face becomes +ve, the other -ve).
- This charge separation creates a small transverse electric field / voltage — the **Hall Voltage**, which balances the magnetic force at equilibrium.

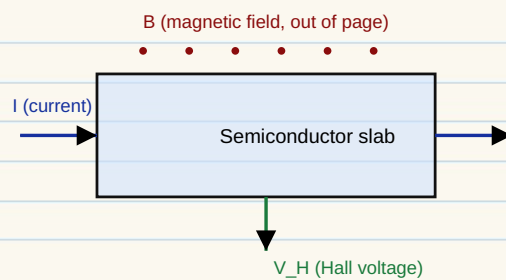


Fig 8.1 — Hall effect setup: I , B and resulting V_H are mutually perpendicular

Important formulas

$$\mathcal{V}_{\mathcal{H}} = (I \cdot B) / (n \cdot e \cdot t)$$

I = current, B = magnetic field, n = carrier concentration, e = electron charge, t = thickness of the slab

$$\mathcal{R}_{\mathcal{H}} = 1 / (n \cdot e)$$

$\mathcal{R}_{\mathcal{H}}$ = Hall coefficient. Sign of $\mathcal{R}_{\mathcal{H}}$ tells the **type of majority carrier**: negative $\mathcal{R}_{\mathcal{H}} \rightarrow$ n-type (electrons); positive $\mathcal{R}_{\mathcal{H}} \rightarrow$ p-type (holes).

Applications of Hall Effect

- To find the **type** of semiconductor (n-type or p-type)
- To find **carrier concentration (n)** and **mobility (μ)** of charge carriers
- Used in **Hall effect sensors** — for measuring magnetic fields, current, and in position/speed sensors (e.g. in cars, brushless motors)

★ **EXAM IMPORTANT**

Derivation of Hall voltage formula, labelled diagram showing I , B and $\mathcal{V}_{\mathcal{H}}$ directions, and "how Hall effect determines carrier type & concentration" are the most frequently asked parts. Numericals using $\mathcal{V}_{\mathcal{H}}$ formula are also common.

🔑 **TRICK TO REMEMBER**

' I , B , $\mathcal{V}$ — all 3 Perpendicular' (like the 3 edges/axes of a box, X-Y-Z) — Current, magnetic field and Hall voltage are always mutually perpendicular to each other.

✔ **QUICK REVISION**

- Hall Effect: $I \perp B$ produces transverse voltage $\mathcal{V}_{\mathcal{H}}$
- $\mathcal{V}_{\mathcal{H}} = IB/ne$; $\mathcal{R}_{\mathcal{H}} = 1/ne$
- Sign of $\mathcal{R}_{\mathcal{H}}$ tells carrier type; used to find n , μ , and B-field sensing

UNIT 3

📁 Final Formula Sheet & Fast Revision

Topic	Key Formula
<i>Kronig-Penney</i>	$\mathcal{P}(\sin \alpha a / \alpha a) + \cos \alpha a = \cos ka$
<i>Diode current</i>	$I = I_0(e^{qV/kT} - 1)$
<i>Mass action law</i>	$n \times p = n_i^2$
<i>Fermi-Dirac function</i>	$f(E) = 1/(1 + e^{(E - E_F)/kT})$
<i>Bloch's theorem</i>	$\Psi(x) = u(x) \cdot e^{ikx}$
<i>Solar cell efficiency</i>	$\eta = (P_{max}/P_{in}) \times 100\%$
<i>Fill factor</i>	$\mathcal{FF} = P_{max}/(V_{oc} \times I_{sc})$
<i>Hall voltage</i>	$V_{H} = IB/(net)$
<i>Hall coefficient</i>	$\mathcal{R}_{H} = 1/(ne)$

🔍 Most Repeated / PYQ Topics (previous year favourites)

1. Derive Kronig-Penney relation & explain origin of energy bands
2. PN junction diode: formation, V-I characteristics, diode equation
3. Differentiate Zener & Avalanche breakdown (table-based question)
4. Fermi level position in intrinsic / N-type / P-type semiconductor (with diagram)
5. Zener diode as voltage regulator (circuit + working)
6. Hall effect: derivation of V_H applications, numericals

🔑 ONE-LINE MEGA MNEMONIC

"King Penney Found Zero Avalanches in Intrinsic Fermi's Solar Zone, Halling loud" → Kronig-Penney, PN diode, Zener, Avalanche, Intrinsic/Extrinsic, Fermi/Bloch, Solar cell, Zener diode, Hall effect — the order of this whole unit!

All the best for your exam! ✨